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Rangkuman praktikum system terdesentralisasi dan terdistribusi

Sinkronisasi Waktu Menggunakan Chrony (NTP) di Ubuntu

The screenshot shows the AWS IAM console 'Edit inbound rules' page for a security group. The table lists four inbound rules:

ID aturan grup keamanan	Jenis	Protokol	Rentang port	Sumber	Deskripsi - opsional	Aksi
sgr-01927254777e2784f	TCP Kustom	TCP	3000	Kustom	0.0.0.0/0	Hapus
sgr-02ac2e802c090fe5e	Semua ICMP - IPv4	ICMP	Semua	Kustom	0.0.0.0/0	Hapus
sgr-09756b4fac08fa674	SSH	TCP	22	Kustom	0.0.0.0/0	Hapus
-	UDP Kustom	UDP	123	IPv4-Di ...	0.0.0.0/0	Hapus

At the bottom, there is a warning message: 'Aturan dengan sumber 0.0.0.0/0 atau ::/0 memungkinkan semua alamat IP mengakses instans Anda. Sebaiknya atur aturan grup keamanan untuk mengizinkan akses hanya dari alamat IP yang diketahui.' Below the warning are buttons for 'Batalkan', 'Pratinjau perubahan', and 'Simpan aturan'.

Pembahasan : Karena kita butuh 3 instans dibuat lagi dan membuat grup keamanan jadi satu untuk server di sever 5 agar edit 1 semua dapat. Kemudian UDP kostum port 123 baik di server atau di client

aws       Amerika Serikat ID Akun: 1386-5493-4658
voclabs/user4457691=irvan.cahya24@students.utdi...

```
# Stop bad estimates upsetting machine clock.
maxupdateskew 100.0

# This directive enables kernel synchronisation (every 11 minutes) of the
# real-time clock. Note that it can't be used along with the 'rtcfile' directive.
rtcsync

# Step the system clock instead of slewing it if the adjustment is larger than
# one second, but only in the first three clock updates.
makestep 1 3

# Get TAI-UTC offset and leap seconds from the system tz database.
# This directive must be commented out when using time sources serving
# leap-smearred time.
leapsectz right/UTC

allow 172.31.24.75

allow 172.31.28.228

driftfile /var/lib/chrony/chrony.drift

log tracking measurements statistics
logdir /var/log/chrony
```

Where Was M-W Next
Forward

i-0d42f28a4cd65cf21 (Vm-server)

PublicIPs: 34.237.146.178 PrivateIPs: 172.31.19.37

```
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
ubuntu@ip-172-31-19-37:~$ sudo nano /etc/chrony/chrony.conf
ubuntu@ip-172-31-19-37:~$ chronyc tracking
Reference ID      : A9FEA97B (169.254.169.123)
Stratum          : 4
Ref time (UTC)   : Thu Oct 30 08:42:27 2025
System time      : 0.000004943 seconds fast of NTP time
Last offset      : +0.000003905 seconds
RMS offset       : 0.000004714 seconds
Frequency        : 7.926 ppm slow
Residual freq    : +0.003 ppm
Skew             : 0.125 ppm
Root delay       : 0.000491915 seconds
Root dispersion  : 0.000339393 seconds
Update interval  : 16.3 seconds
Leap status      : Normal
ubuntu@ip-172-31-19-37:~$
```

Pembahasan : pertama **server** dikondisikan untuk install beberapa hal agar bisa masuk sudo nano setelah itu ip client di allow seperti diatas dan beberapa setting ditambahkan agar bisa jalan chronyc tracking untuk cek

```
GNU nano 7.2 /etc/chrony/chrony.conf *
# Welcome to the chrony configuration file. See chrony.conf(5) for more
# information about usable directives.

# Include configuration files found in /etc/chrony/conf.d.
confdir /etc/chrony/conf.d

# This will use (up to):
# - 4 sources from ntp.ubuntu.com which some are ipv6 enabled
# - 2 sources from 2.ubuntu.pool.ntp.org which is ipv6 enabled as well
# - 1 source from [01].ubuntu.pool.ntp.org each (ipv4 only atm)
# This means by default, up to 6 dual-stack and up to 2 additional IPv4-only
# sources will be used.
# At the same time it retains some protection against one of the entries being
# down (compare to just using one of the lines). See (LP: #1754358) for the
# discussion.
#
# About using servers from the NTP Pool Project in general see (LP: #104525).
# Approved by Ubuntu Technical Board on 2011-02-08.
# See http://www.pool.ntp.org/join.html for more information.
#pool ntp.ubuntu.com iburst maxsources 4
#pool 0.ubuntu.pool.ntp.org iburst maxsources 1
#pool 1.ubuntu.pool.ntp.org iburst maxsources 1
#pool 2.ubuntu.pool.ntp.org iburst maxsources 2

server 172.31.19.37 iburst prefer

# Use time sources from DHCP.
sourcedir /run/chrony-dhcp

# Use NTP sources found in /etc/chrony/sources.d.

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Pembahasan : vm client dikonfigurasi agar menjadikan vm server menjadi server untuk clietn

```

ubuntu@ip-172-31-24-75:~$ sudo nano /etc/chrony/chrony.conf
ubuntu@ip-172-31-24-75:~$ sudo systemctl restart chronyd
Warning: The unit file, source configuration file or drop-ins of chronyd.service changed on
disk. Run 'systemctl daemon-reload' to reload units.
ubuntu@ip-172-31-24-75:~$ chronyc sources -v

.-- Source mode  '^' = server, '=' = peer, '#' = local clock.
/ .- Source state '*' = current best, '+' = combined, '-' = not combined,
| /           'x' = may be in error, '~' = too variable, '?' = unusable.
||
||           Reachability register (octal) --.          | xxxxx [ yyyy ] +/- zzzz
||           Log2(Polling interval) --.          |          | xxxxx = adjusted offset,
||           \          |          |          |          | yyyy = measured offset,
||           |          |          |          |          | zzzz = estimated error.
||           \          |          |          |          |
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^* 169.254.169.123           3  4   377    7    -13us[ -34us] +/-  541us
^? ip-172-31-19-37.ec2.inte> 0  7     0    -    +0ns[ +0ns] +/-    0ns
ubuntu@ip-172-31-24-75:~$

```

i-000a8ff1c561f54d3 (Vm-client)

```

.-- Source mode  '^' = server, '=' = peer, '#' = local clock.
/ .- Source state '*' = current best, '+' = combined, '-' = not combined,
| /           'x' = may be in error, '~' = too variable, '?' = unusable.
||
||           Reachability register (octal) --.          | xxxxx [ yyyy ] +/- zzzz
||           Log2(Polling interval) --.          |          | xxxxx = adjusted offset,
||           \          |          |          |          | yyyy = measured offset,
||           |          |          |          |          | zzzz = estimated error.
||           \          |          |          |          |
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^? 169.254.169.123           0  4     0    -    +0ns[ +0ns] +/-    0ns
^* ip-172-31-19-37.ec2.inte>  4  6    17   11  +1626ns[+7231ns] +/-  813us
ubuntu@ip-172-31-28-228:~$ chronyc tracking
Reference ID      : AC1F1325 (ip-172-31-19-37.ec2.internal)
Stratum          : 5
Ref time (UTC)   : Thu Oct 30 09:16:39 2025
System time      : 0.000000006 seconds slow of NTP time
Last offset      : +0.000005605 seconds
RMS offset       : 0.000005605 seconds
Frequency        : 2.910 ppm slow
Residual freq    : +1.965 ppm
Skew             : 2.974 ppm
Root delay       : 0.000740165 seconds
Root dispersion  : 0.000813881 seconds
Update interval  : 2.0 seconds
Leap status      : Normal
ubuntu@ip-172-31-28-228:~$

```

Pembahasan : memblokir server dari amerika dan membuat vm server menjadi server nyata

```
ubuntu@ip-172-31-19-37:~$ date
Thu Oct 30 09:22:37 UTC 2025
ubuntu@ip-172-31-19-37:~$
```

i-0d42f28a4cd65cf21 (Vm-server)

PublicIPs: 34.237.146.178 PrivateIPs: 172.31.19.37

```
Thu Oct 30 09:20:21 UTC 2025
ubuntu@ip-172-31-28-228:~$
```

i-053840aae81b5b7ce (vm-client2)

PublicIPs: 54.221.62.7 PrivateIPs: 172.31.28.228

```
Reference ID : AC1F1325 (ip-172-31-19-37.ec2.internal)
Stratum : 5
Ref time (UTC) : Thu Oct 30 09:16:39 2025
System time : 0.000000006 seconds slow of NTP time
Last offset : +0.000005605 seconds
RMS offset : 0.000005605 seconds
Frequency : 2.910 ppm slow
Residual freq : +1.965 ppm
Skew : 2.974 ppm
Root delay : 0.000740165 seconds
Root dispersion : 0.000813881 seconds
Update interval : 2.0 seconds
Leap status : Normal
ubuntu@ip-172-31-28-228:~$ sudo date -s "-2 minutes"
Thu Oct 30 09:20:00 UTC 2025
ubuntu@ip-172-31-28-228:~$ date
Thu Oct 30 09:20:21 UTC 2025
ubuntu@ip-172-31-28-228:~$ sudo chronyc makestep
200 OK
ubuntu@ip-172-31-28-228:~$ chronyc tracking
Reference ID : AC1F1325 (ip-172-31-19-37.ec2.internal)
Stratum : 5
Ref time (UTC) : Thu Oct 30 09:24:12 2025
System time : 119.771057129 seconds slow of NTP time
Last offset : -120.000061035 seconds
RMS offset : 37.947349548 seconds
Frequency : 3.293 ppm slow
Residual freq : -0.376 ppm
Skew : 0.399 ppm
Root delay : 0.000709264 seconds
Root dispersion : 0.000411993 seconds
Update interval : 0.0 seconds
Leap status : Normal
ubuntu@ip-172-31-28-228:~$
```

i-053840aae81b5b7ce (vm-client2)

PublicIPs: 54.221.62.7 PrivateIPs: 172.31.28.228

Pembahasan : membuat waktu delay antara client dan sever sebagai bukti sudah berjalan dengan lancar kemudain sinkronisasi dikembalikan seperti normal

Kesimpulan : membuat server dan client tersambung di aws